

Hadoop MapReduce Word Count on Multiple Files

1. Objective

To create an HDFS directory, upload multiple text files, perform Word Count using Hadoop MapReduce on all files collectively, and identify the top 3 most frequent words from the generated output.

2. Procedure

Step 1: Create a Directory in HDFS

```
hdfs dfs -mkdir /my_data_directory
```

Step 2: Create Three Text Files Locally

```
echo "This is the content of file1.txt" > file1.txt  
echo "This is the content of file2.txt" > file2.txt  
echo "This is the content of file3.txt" > file3.txt
```

Step 3: Upload the Files to HDFS

```
hdfs dfs -put file1.txt /my_data_directory/  
hdfs dfs -put file2.txt /my_data_directory/  
hdfs dfs -put file3.txt /my_data_directory/
```

Step 4: Verify the Uploaded Files

List Files in the Directory

```
hdfs dfs -ls /my_data_directory
```

View File Contents

```
hdfs dfs -cat /my_data_directory/file1.txt  
hdfs dfs -cat /my_data_directory/file2.txt  
hdfs dfs -cat /my_data_directory/file3.txt
```

Step 5: Execute the Word Count Program

```
hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar  
wordcount /my_data_directory /wordcount_output
```

Step 6: View the Word Count Results

```
hdfs dfs -cat /wordcount_output/part-r-000000
```

Step 7: Find the Top 3 Most Frequent Words

```
hdfs dfs -cat /wordcount_output/part-r-000000 | sort -t"$\t" -k2 -nr | head -3
```

3. Source Code

Create HDFS Directory

```
hdfs dfs -mkdir /my_data_directory
```

Create Input Files

```
echo "This is the content of file1.txt" > file1.txt  
echo "This is the content of file2.txt" > file2.txt  
echo "This is the content of file3.txt" > file3.txt
```

Upload Files to HDFS

```
hdfs dfs -put file1.txt /my_data_directory/  
hdfs dfs -put file2.txt /my_data_directory/  
hdfs dfs -put file3.txt /my_data_directory/
```

Verify Uploaded Files

```
hdfs dfs -ls /my_data_directory
```

```
hdfs dfs -cat /my_data_directory/file1.txt  
hdfs dfs -cat /my_data_directory/file2.txt  
hdfs dfs -cat /my_data_directory/file3.txt
```

Execute Word Count

```
hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar  
wordcount /my_data_directory /wordcount_output
```

View Results

```
hdfs dfs -cat /wordcount_output/part-r-000000
```

Get Top 3 Most Frequent Words

```
hdfs dfs -cat /wordcount_output/part-r-00000 | sort -t$'\t' -k2 -nr | head -3
```

4. Screenshots

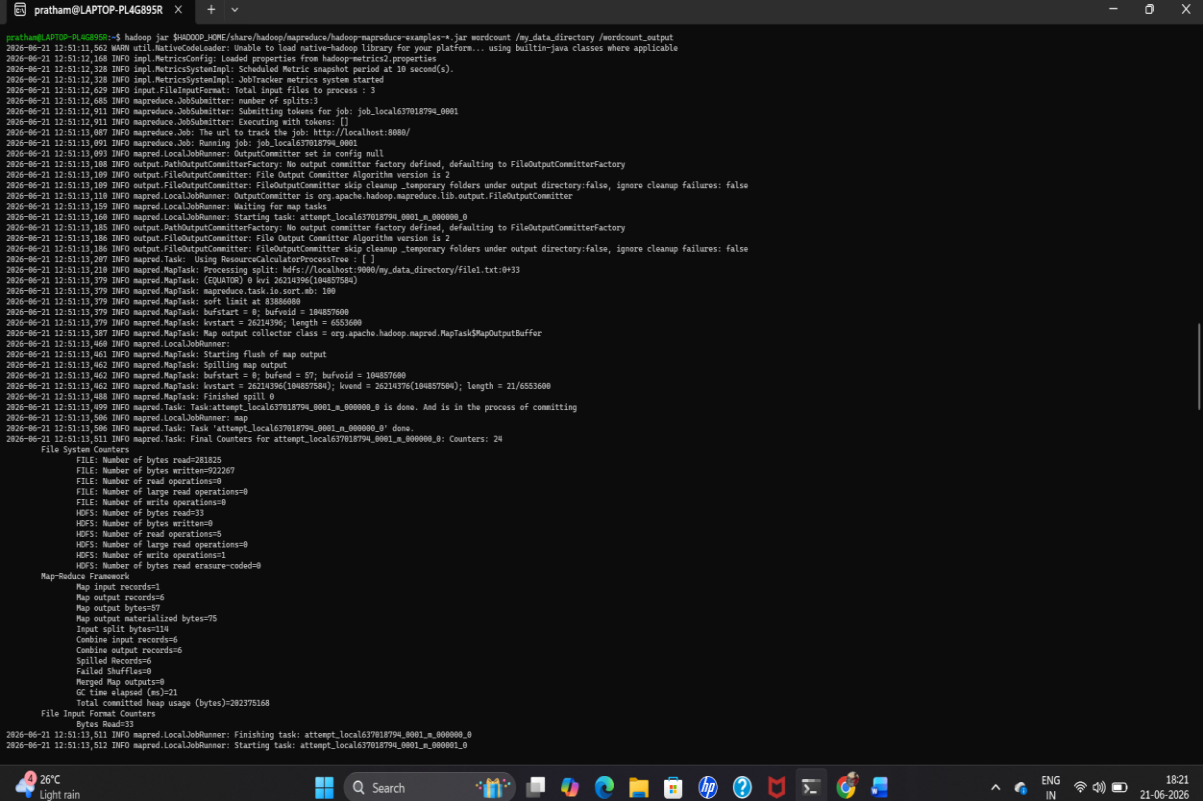
```
pratham@LAPTOP-PL4G895R:~$ hdfs dfs -mkdir /my_data_directory

pratham@LAPTOP-PL4G895R:~$ echo "This is the content of file1.txt" > file1.txt
echo "This is the content of file2.txt" > file2.txt
echo "This is the content of file3.txt" > file3.txt

pratham@LAPTOP-PL4G895R:~$ hdfs dfs -put file1.txt /my_data_directory/
hdfs dfs -put file2.txt /my_data_directory/
hdfs dfs -put file3.txt /my_data_directory/

pratham@LAPTOP-PL4G895R:~$ hdfs dfs -ls /my_data_directory
2026-06-21 12:50:01,483 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 3 items
-rw-r--r-- 1 pratham supergroup 33 2026-06-21 12:49 /my_data_directory/file1.txt
-rw-r--r-- 1 pratham supergroup 33 2026-06-21 12:49 /my_data_directory/file2.txt
-rw-r--r-- 1 pratham supergroup 33 2026-06-21 12:49 /my_data_directory/file3.txt

pratham@LAPTOP-PL4G895R:~$ hdfs dfs -cat /my_data_directory/file1.txt
hdfs dfs -cat /my_data_directory/file2.txt
hdfs dfs -cat /my_data_directory/file3.txt
2026-06-21 12:50:38,461 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
This is the content of file1.txt
2026-06-21 12:50:39,812 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
This is the content of file2.txt
2026-06-21 12:50:41,162 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
This is the content of file3.txt
```



```
pratham@LAPTOP-PL4G895R X + v
2026-06-21 12:51:13,513 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-21 12:51:13,513 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-21 12:51:13,513 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-06-21 12:51:13,513 INFO mapred.Task: Using ResourceCalculatorProcessTree: [ ]
2026-06-21 12:51:13,513 INFO mapred.Task: https://localhost:9000/my_data_directory/file2.txt:0+33
2026-06-21 12:51:13,536 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-06-21 12:51:13,536 INFO mapred.MapTask: soft limit at: 1386868
2026-06-21 12:51:13,537 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-06-21 12:51:13,537 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-06-21 12:51:13,539 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-06-21 12:51:13,560 INFO mapred.LocalJobRunner: map
2026-06-21 12:51:13,560 INFO mapred.MapTask: Starting flush of map output
2026-06-21 12:51:13,560 INFO mapred.MapTask: Spilling map output
2026-06-21 12:51:13,560 INFO mapred.MapTask: bufstart = 0; bufend = 57; bufvoid = 104857600
2026-06-21 12:51:13,570 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214376(104857584); length = 21/6553600
2026-06-21 12:51:13,571 INFO mapred.MapTask: Finished spill 0
2026-06-21 12:51:13,576 INFO mapred.Task: Task:attempt_local637018794_0001_m_000001_0 is done. And is in the process of committing
2026-06-21 12:51:13,579 INFO mapred.Task: Task:attempt_local637018794_0001_m_000001_0' done
2026-06-21 12:51:13,580 INFO mapred.Task: Final Counters for attempt_local637018794_0001_m_000001_0: Counters: 24
File System Counters
FILE: Number of bytes read=282186
FILE: Number of bytes written=922374
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=66
HDFS: Number of bytes written=0
HDFS: Number of read operations=7
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=1
Map output records=6
Map output bytes=57
Map output materialized bytes=75
Input split bytes=134
Combine input records=6
Combine output records=6
Spilled Records=6
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=18
Total committed heap usage (bytes)=202375168
File Input Format Counters
Bytes Read=33
2026-06-21 12:51:13,580 INFO mapred.LocalJobRunner: Finishing task: attempt_local637018794_0001_m_000001_0
2026-06-21 12:51:13,581 INFO mapred.LocalJobRunner: Starting task: attempt_local637018794_0001_m_000002_0
2026-06-21 12:51:13,581 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-21 12:51:13,581 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-21 12:51:13,581 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-06-21 12:51:13,582 INFO mapred.Task: Using ResourceCalculatorProcessTree: [ ]
2026-06-21 12:51:13,587 INFO mapred.MapTask: Processing split: https://localhost:9000/my_data_directory/file3.txt:0+33
2026-06-21 12:51:13,639 INFO mapred.MapTask: (CQAO) 0 kv: 26214376(104857584)
2026-06-21 12:51:13,639 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-06-21 12:51:13,639 INFO mapred.MapTask: soft limit at: 8386868
2026-06-21 12:51:13,639 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-06-21 12:51:13,639 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-06-21 12:51:13,640 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-06-21 12:51:13,653 INFO mapred.LocalJobRunner: map
2026-06-21 12:51:13,653 INFO mapred.MapTask: Starting flush of map output
2026-06-21 12:51:13,653 INFO mapred.MapTask: Spilling map output
2026-06-21 12:51:13,653 INFO mapred.MapTask: bufstart = 0; bufend = 57; bufvoid = 104857600
2026-06-21 12:51:13,662 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214376(104857584); length = 21/6553600
2026-06-21 12:51:13,671 INFO mapred.Task: Task:attempt_local637018794_0001_m_000002_0 is done. And is in the process of committing

2026-06-21 12:51:13,674 INFO mapred.LocalJobRunner: map
2026-06-21 12:51:13,674 INFO mapred.Task: Task:attempt_local637018794_0001_m_000002_0' done
2026-06-21 12:51:13,674 INFO mapred.Task: Final Counters for attempt_local637018794_0001_m_000002_0: Counters: 24
File System Counters
FILE: Number of bytes read=282547
FILE: Number of bytes written=922481
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=99
HDFS: Number of bytes written=0
HDFS: Number of read operations=9
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=1
Map output records=6
Map output bytes=57
Map output materialized bytes=75
Input split bytes=134
Combine input records=6
Combine output records=6
Spilled Records=6
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=15
Total committed heap usage (bytes)=234818024
File Input Format Counters
Bytes Read=33
2026-06-21 12:51:13,675 INFO mapred.LocalJobRunner: Finishing task: attempt_local637018794_0001_m_000002_0
2026-06-21 12:51:13,675 INFO mapred.LocalJobRunner: map task execution complete.
2026-06-21 12:51:13,679 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-06-21 12:51:13,679 INFO mapred.LocalJobRunner: Starting task: attempt_local637018794_0001_r_000000_0
2026-06-21 12:51:13,687 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-21 12:51:13,687 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-21 12:51:13,687 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-06-21 12:51:13,688 INFO mapred.Task: Using ResourceCalculatorProcessTree: [ ]
2026-06-21 12:51:13,693 INFO mapred.ReduceTask: org.apache.hadoop.mapreduce.task.reduce.Shuffle@4da4db0
2026-06-21 12:51:13,693 INFO mapred.ReduceTask: org.apache.hadoop.mapreduce.task.reduce.Shuffle@4da4db0
2026-06-21 12:51:13,693 INFO mapred.ReduceTask: org.apache.hadoop.mapreduce.task.reduce.Shuffle@4da4db0
2026-06-21 12:51:13,712 INFO reduce.MergeManagerImpl: MergeManager: memoryLimit=68849976, maxSingleShuffleInit=172223704, mergeThreshold=45486688, insertFactor=10, memToMemMergeOutputsThreshold=10
2026-06-21 12:51:13,715 INFO reduce.EventFetcher: attempt_local637018794_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-06-21 12:51:13,762 INFO reduce.LocalFetcher: localfetcher1 about to shuffle output of map attempt_local637018794_0001_m_000001_0 decomp: 71 len: 75 to MEMORY
2026-06-21 12:51:13,765 INFO reduce.InMemoryOutput: Read 71 bytes from map-output for attempt_local637018794_0001_m_000001_0
2026-06-21 12:51:13,767 INFO reduce.MergeManagerImpl: classInMemoryFile -> map-output of size: 71, inMemoryMapOutputs.size() -> 3, commitMemory -> 0, useMemory -> 71
2026-06-21 12:51:13,767 INFO reduce.LocalFetcher: localfetcher1 about to shuffle output of map attempt_local637018794_0001_m_000001_0 decomp: 71 len: 75 to MEMORY
2026-06-21 12:51:13,768 INFO reduce.InMemoryOutput: Read 71 bytes from map-output for attempt_local637018794_0001_m_000001_0
2026-06-21 12:51:13,768 INFO reduce.MergeManagerImpl: classInMemoryFile -> map-output of size: 71, inMemoryMapOutputs.size() -> 2, commitMemory -> 71, useMemory -> 242
2026-06-21 12:51:13,768 INFO reduce.LocalFetcher: localfetcher1 about to shuffle output of map attempt_local637018794_0001_m_000000_0 decomp: 71 len: 75 to MEMORY
2026-06-21 12:51:13,768 INFO reduce.InMemoryOutput: Read 71 bytes from map-output for attempt_local637018794_0001_m_000000_0
2026-06-21 12:51:13,768 INFO reduce.MergeManagerImpl: classInMemoryFile -> map-output of size: 71, inMemoryMapOutputs.size() -> 3, commitMemory -> 142, useMemory -> 213
2026-06-21 12:51:13,769 INFO reduce.EventFetcher: EventFetcher is interrupted. Returning
2026-06-21 12:51:13,769 INFO mapred.LocalJobRunner: 0 / 3 copied
2026-06-21 12:51:13,769 INFO reduce.MergeManagerImpl: finalMerge called with 3 in-memory map-outputs and 0 on-disk map-outputs
2026-06-21 12:51:13,773 INFO mapred.Namer: Merging 3 sorted segments
2026-06-21 12:51:13,773 INFO reduce.MergeManagerImpl: MergeManager: Done to the last merge-pass, with 3 segments left of total size: 192 bytes
2026-06-21 12:51:13,780 INFO reduce.MergeManagerImpl: Merged 3 segments, 213 bytes to disk to satisfy reduce memory limit
2026-06-21 12:51:13,780 INFO reduce.MergeManagerImpl: Merging 3 files, 213 bytes from disk
2026-06-21 12:51:13,781 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-06-21 12:51:13,781 INFO mapred.Namer: Merging 3 sorted segments
2026-06-21 12:51:13,781 INFO mapred.Namer: Done to the last merge-pass, with 1 segments left of total size: 202 bytes
2026-06-21 12:51:13,782 INFO mapred.LocalJobRunner: 1 / 3 copied
2026-06-21 12:51:13,812 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-06-21 12:51:13,860 INFO mapred.Task: Task:attempt_local637018794_0001_r_000000_0 is done. And is in the process of committing
2026-06-21 12:51:13,860 INFO mapred.LocalJobRunner: 1 / 3 copied
2026-06-21 12:51:13,860 INFO mapred.Task: Task:attempt_local637018794_0001_r_000000_0 is allowed to commit now
2026-06-21 12:51:13,869 INFO output.FileOutputCommitter: Saved output of task 'attempt_local637018794_0001_r_000000_0' to https://localhost:9000/wordcount-output
2026-06-21 12:51:13,883 INFO mapred.LocalJobRunner: reduce r reduce

2026-06-21 12:51:13,674 INFO mapred.LocalJobRunner: map
2026-06-21 12:51:13,674 INFO mapred.Task: Task:attempt_local637018794_0001_m_000002_0' done
2026-06-21 12:51:13,674 INFO mapred.Task: Final Counters for attempt_local637018794_0001_m_000002_0: Counters: 24
File System Counters
FILE: Number of bytes read=282547
FILE: Number of bytes written=922481
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=99
HDFS: Number of bytes written=0
HDFS: Number of read operations=9
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=1
Map output records=6
Map output bytes=57
Map output materialized bytes=75
Input split bytes=134
Combine input records=6
Combine output records=6
Spilled Records=6
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=15
Total committed heap usage (bytes)=234818024
File Input Format Counters
Bytes Read=33
2026-06-21 12:51:13,675 INFO mapred.LocalJobRunner: Finishing task: attempt_local637018794_0001_m_000002_0
2026-06-21 12:51:13,675 INFO mapred.LocalJobRunner: map task execution complete.
2026-06-21 12:51:13,679 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-06-21 12:51:13,679 INFO mapred.LocalJobRunner: Starting task: attempt_local637018794_0001_r_000000_0
2026-06-21 12:51:13,687 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-06-21 12:51:13,687 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-06-21 12:51:13,687 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-06-21 12:51:13,688 INFO mapred.Task: Using ResourceCalculatorProcessTree: [ ]
2026-06-21 12:51:13,693 INFO mapred.ReduceTask: org.apache.hadoop.mapreduce.task.reduce.Shuffle@4da4db0
2026-06-21 12:51:13,693 INFO mapred.ReduceTask: org.apache.hadoop.mapreduce.task.reduce.Shuffle@4da4db0
2026-06-21 12:51:13,693 INFO mapred.ReduceTask: org.apache.hadoop.mapreduce.task.reduce.Shuffle@4da4db0
2026-06-21 12:51:13,712 INFO reduce.MergeManagerImpl: MergeManager: memoryLimit=68849976, maxSingleShuffleInit=172223704, mergeThreshold=45486688, insertFactor=10, memToMemMergeOutputsThreshold=10
2026-06-21 12:51:13,715 INFO reduce.EventFetcher: attempt_local637018794_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-06-21 12:51:13,762 INFO reduce.LocalFetcher: localfetcher1 about to shuffle output of map attempt_local637018794_0001_m_000001_0 decomp: 71 len: 75 to MEMORY
2026-06-21 12:51:13,765 INFO reduce.InMemoryOutput: Read 71 bytes from map-output for attempt_local637018794_0001_m_000001_0
2026-06-21 12:51:13,767 INFO reduce.MergeManagerImpl: classInMemoryFile -> map-output of size: 71, inMemoryMapOutputs.size() -> 3, commitMemory -> 0, useMemory -> 71
2026-06-21 12:51:13,767 INFO reduce.LocalFetcher: localfetcher1 about to shuffle output of map attempt_local637018794_0001_m_000001_0 decomp: 71 len: 75 to MEMORY
2026-06-21 12:51:13,768 INFO reduce.InMemoryOutput: Read 71 bytes from map-output for attempt_local637018794_0001_m_000001_0
2026-06-21 12:51:13,768 INFO reduce.MergeManagerImpl: classInMemoryFile -> map-output of size: 71, inMemoryMapOutputs.size() -> 2, commitMemory -> 71, useMemory -> 242
2026-06-21 12:51:13,768 INFO reduce.LocalFetcher: localfetcher1 about to shuffle output of map attempt_local637018794_0001_m_000000_0 decomp: 71 len: 75 to MEMORY
2026-06-21 12:51:13,768 INFO reduce.InMemoryOutput: Read 71 bytes from map-output for attempt_local637018794_0001_m_000000_0
2026-06-21 12:51:13,768 INFO reduce.MergeManagerImpl: classInMemoryFile -> map-output of size: 71, inMemoryMapOutputs.size() -> 3, commitMemory -> 142, useMemory -> 213
2026-06-21 12:51:13,769 INFO reduce.EventFetcher: EventFetcher is interrupted. Returning
2026-06-21 12:51:13,769 INFO mapred.LocalJobRunner: 0 / 3 copied
2026-06-21 12:51:13,769 INFO reduce.MergeManagerImpl: finalMerge called with 3 in-memory map-outputs and 0 on-disk map-outputs
2026-06-21 12:51:13,773 INFO mapred.Namer: Merging 3 sorted segments
2026-06-21 12:51:13,773 INFO reduce.MergeManagerImpl: MergeManager: Done to the last merge-pass, with 3 segments left of total size: 192 bytes
2026-06-21 12:51:13,780 INFO reduce.MergeManagerImpl: Merged 3 segments, 213 bytes to disk to satisfy reduce memory limit
2026-06-21 12:51:13,780 INFO reduce.MergeManagerImpl: Merging 3 files, 213 bytes from disk
2026-06-21 12:51:13,781 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-06-21 12:51:13,781 INFO mapred.Namer: Merging 3 sorted segments
2026-06-21 12:51:13,781 INFO mapred.Namer: Done to the last merge-pass, with 1 segments left of total size: 202 bytes
2026-06-21 12:51:13,782 INFO mapred.LocalJobRunner: 1 / 3 copied
2026-06-21 12:51:13,812 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-06-21 12:51:13,860 INFO mapred.Task: Task:attempt_local637018794_0001_r_000000_0 is done. And is in the process of committing
2026-06-21 12:51:13,860 INFO mapred.LocalJobRunner: 1 / 3 copied
2026-06-21 12:51:13,860 INFO mapred.Task: Task:attempt_local637018794_0001_r_000000_0 is allowed to commit now
2026-06-21 12:51:13,869 INFO output.FileOutputCommitter: Saved output of task 'attempt_local637018794_0001_r_000000_0' to https://localhost:9000/wordcount-output
2026-06-21 12:51:13,883 INFO mapred.LocalJobRunner: reduce r reduce
```

```
pratham@LAPTOP-PL4G89SR: ~$ cat /dev/null > /dev/null &
2026-06-21 12:51:13,883 INFO mapred.Task: Task 'attempt_local67918794_0001_r_000000_0' done
2026-06-21 12:51:13,885 INFO mapred.Task: Final Counters for attempt_local67918794_0001_r_000000_0: Counters: 38
File System Counters
  FILE: Number of bytes read=283801
  FILE: Number of bytes written=922698
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=99
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=3
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Combine input records=9
  Combine output records=0
  Reduce input groups=8
  Reduce shuffle bytes=225
  Reduce input records=18
  Reduce output records=8
  Spilled Records=18
  Shuffled Maps=3
  Failed Shuffles=0
  Merged Map outputs=3
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=234881824
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDCOUNT=0
File Output Format Counters
  Bytes Written=69
2026-06-21 12:51:13,885 INFO mapred.LocalJobRunner: Finishing task: attempt_local67918794_0001_r_000000_0
2026-06-21 12:51:13,885 INFO mapred.LocalJobRunner: reduce task executor complete
2026-06-21 12:51:14,099 INFO mapreduce.Job: Job job_local67918794_0001 running in user mode : false
2026-06-21 12:51:14,100 INFO mapreduce.Job: map 100% reduce 100%
2026-06-21 12:51:14,162 INFO mapreduce.Job: Job job_local67918794_0001 completed successfully
2026-06-21 12:51:14,111 INFO mapreduce.Job: Counters: 36
File System Counters
  FILE: Number of bytes read=1129639
  FILE: Number of bytes written=3689816
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=297
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=35
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=6
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=3
  Map output records=18
  Map output bytes=171
  Map output materialized bytes=225
  Input split bytes=92
  Combine input records=18
  Combine output records=18
  Reduce input groups=8
  Reduce shuffle bytes=225
  Reduce input records=18
  Reduce output records=8
  Spilled Records=36
  Shuffled Maps=3
  Failed Shuffles=0
  Merged Map outputs=3
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=879812384
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDCOUNT=0
File Input Format Counters
  Bytes Read=99
File Output Format Counters
  Bytes Written=69
pratham@LAPTOP-PL4G89SR: ~$
```

```
pratham@LAPTOP-PL4G89SR: ~$ cat /dev/null > /dev/null &
2026-06-21 12:51:13,883 INFO mapred.Task: Task 'attempt_local67918794_0001_r_000000_0' done
2026-06-21 12:51:13,885 INFO mapred.Task: Final Counters for attempt_local67918794_0001_r_000000_0: Counters: 38
File System Counters
  FILE: Number of bytes read=283801
  FILE: Number of bytes written=922698
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=99
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=3
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Combine input records=9
  Combine output records=0
  Reduce input groups=8
  Reduce shuffle bytes=225
  Reduce input records=18
  Reduce output records=8
  Spilled Records=18
  Shuffled Maps=3
  Failed Shuffles=0
  Merged Map outputs=3
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=234881824
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDCOUNT=0
File Output Format Counters
  Bytes Written=69
2026-06-21 12:51:13,885 INFO mapred.LocalJobRunner: Finishing task: attempt_local67918794_0001_r_000000_0
2026-06-21 12:51:13,885 INFO mapred.LocalJobRunner: reduce task executor complete
2026-06-21 12:51:14,099 INFO mapreduce.Job: Job job_local67918794_0001 running in user mode : false
2026-06-21 12:51:14,100 INFO mapreduce.Job: map 100% reduce 100%
2026-06-21 12:51:14,162 INFO mapreduce.Job: Job job_local67918794_0001 completed successfully
2026-06-21 12:51:14,111 INFO mapreduce.Job: Counters: 36
File System Counters
  FILE: Number of bytes read=1129639
  FILE: Number of bytes written=3689816
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=297
  HDFS: Number of bytes written=69
  HDFS: Number of read operations=35
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=6
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=3
  Map output records=18
  Map output bytes=171
  Map output materialized bytes=225
  Input split bytes=92
  Combine input records=18
  Combine output records=18
  Reduce input groups=8
  Reduce shuffle bytes=225
  Reduce input records=18
  Reduce output records=8
  Spilled Records=36
  Shuffled Maps=3
  Failed Shuffles=0
  Merged Map outputs=3
  GC time elapsed (ms)=0
  Total committed heap usage (bytes)=879812384
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDCOUNT=0
File Input Format Counters
  Bytes Read=99
File Output Format Counters
  Bytes Written=69
pratham@LAPTOP-PL4G89SR: ~$
```

```
pratham@LAPTOP-PL4G89SR: ~$ hdfs dfs -cat /wordcount_output/part-r-000000
2026-06-21 12:53:06,811 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
This      3
content  3
file1.txt 1
file2.txt 1
file3.txt 1
is        3
of        3
the       3
```

```
pratham@LAPTOP-PL4G89SR: ~$ hdfs dfs -cat /wordcount_output/part-r-000000 | sort -t'$t' -k2 -nr | head -3
2026-06-21 12:53:43,692 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
the       3
of        3
is        3
```

5. Conclusion

The experiment successfully demonstrated Hadoop MapReduce processing on multiple input files stored in HDFS. Three text files were uploaded into an HDFS directory and processed collectively using the Word Count application. The generated output provided the frequency of each word across all files, and Linux commands were used to identify the top three most frequently occurring words. This experiment illustrates Hadoop's capability to process data distributed across multiple files efficiently.